

Lab 3 – Component Modelling & Architectural Pattern Selection

Problem Statement #29 — Municipal Infrastructure Damage Reporting App

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Scenario Context: Citizens report potholes, broken streetlights, and pipeline leaks using geotagged photos, and the app automatically dispatches a ticket to the appropriate municipal department. The architecture addresses EXIF/GPS extraction and offline drafting with auto-sync.

1. Architecture Selection

We chose Microservices Architecture for the Municipal Infrastructure Damage Reporting App.

2. Reason 1 — Department Separation

Road Infrastructure, Water, and Electrical departments are separate downstream consumers of a ticket. Keeping ticket dispatch as its own service means routing rules for one department can be changed without tightly coupling the other department workflows.

3. Reason 2 — Bursty Workload

Photo upload and EXIF/GPS extraction can experience sudden spikes after events such as heavy rain, while the Engineer Portal may have steadier traffic. Separating these responsibilities allows the report-intake and photo-processing service to scale independently.

4. Security Advantage

The Citizen Reporter App does not access the Ticket Data Store directly. It communicates through the Report Intake service's public API, reducing direct exposure of ticket and department data and keeping dispatch/data access behind service boundaries.

5. Performance Benefit

The Report Intake & Photo Processing Service can be scaled horizontally during peak reporting periods without scaling every other service. This avoids over-provisioning the rest of the system while maintaining responsiveness for photo and GPS processing

PS-29 — Municipal Infrastructure Damage Reporting App
UML Component Diagram

